FYP PROGRESS REPORT · PHASE 2

Siamese Swap-Test QNN for Copy-Move Forgery Detection

Replacing the v1 per-patch classifier with a pairwise-matching quantum architecture. This page tracks what's been verified, what broke, what was fixed, and what's still open.

- architecture bug found & fixed
- 3 training runs completed
- real-image testing: in progress
- rotation-aware filtering: not started
- visualization tool: not started

01 **Real CoMoFoD dataset wired up, category labels corrected**

02 **Training loss flat → traced to a dead gradient**

03 **Root cause proven: tied weights invisible to swap test**

04 **Fixed with data re-uploading; gradient confirmed real**

05 **Full run: 10 epochs, entire clean pool, 60k pairs**

06 **Real-image F1/IoU benchmark — in progress**

07 **Rotation-aware offset voting — not started**

§1 What went wrong

A 5-epoch training smoke test showed a loss that never decreased. Rather than assume it just needed more epochs, the gradient was measured directly.

• CRITICAL — ROOT CAUSE

The tied weights were mathematically invisible to the swap test

Both quantum registers were embedded with fixed data, then the *same* unitary U(w) was applied to both before the swap test read out their overlap. For any unitary, that overlap reduces algebraically to the raw input overlap — the weights cancel out completely, everywhere in parameter space:

$$|\langle U(w)f_a \mid U(w)f_b\rangle|^2 = |\langle f_a| U^\dagger(w)U(w) |f_b\rangle|^2 = |\langle f_a|f_b\rangle|^2 \quad \text{for ANY } U(w)$$

Confirmed empirically two ways: the measured gradient was ~6.5×10^{-16} (zero to machine precision) at every qubit count tested, and five independent random weight settings produced the *exact same* similarity score — matching the raw classical cosine similarity of the input features, bit for bit.

GRADIENT NORM, PRE-FIX

6.5×10^{-16}

zero to machine precision

WEIGHT MOVEMENT OVER 5 EPOCHS

$\sim 3 \times 10^{-9}$

floating-point noise only

SIMILARITY ACROSS 5 RANDOM WEIGHT DRAWS

0.083549

identical, every time

§2 The fix

✓ RESOLVED — DATA RE-UPLOADING

Weights and data now interleave inside the same gates

Instead of a fixed embedding followed by a data-independent weight block, the trainable weights and each register's own classical features are now combined additively inside the same rotation gates, repeated across layers. Because register A and register B carry different data at every layer, there's no shared data-independent unitary left for the swap test to cancel against. Tied weights and the swap-test readout — both flagged as deliberate — are unchanged.

GRADIENT NORM

~~~0~~ → **0.027**

WEIGHT MOVEMENT (PER RUN)

~~3e-9~~ → **0.46**

HELD-OUT SEPARATION, FULL RUN

0.111 → 0.189

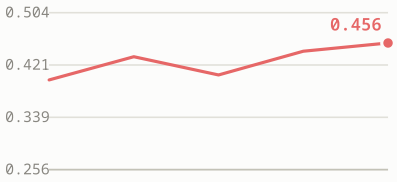
monotonic across 10 epochs
~~~

§3 Training runs

Three runs, same fixed architecture, increasing scale. Raw per-epoch loss stays noisy in all of them — each epoch redraws a random image sample, so batch noise dominates at these run sizes. The shared y-axis below makes the level shift visible: pre-fix hovers near 0.40–0.46 with no possible mechanism to move (the gradient was zero); post-fix runs settle lower, around 0.30–0.35.

Pre-fix (broken)

5 epochs · gradient ≈ 0



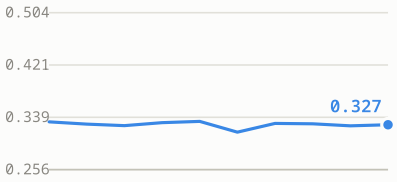
Medium run

15 epochs · 50 img/epoch



Full run

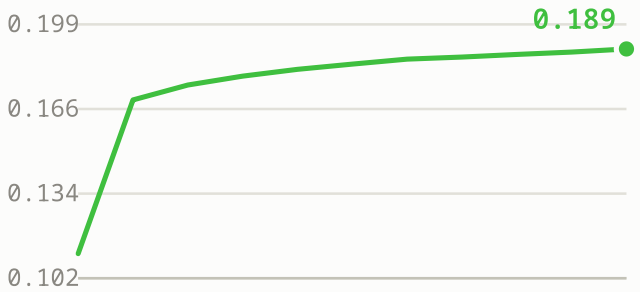
10 epochs · 200 img/epoch



The cleaner signal is **held-out separation** — positive vs. negative pair scores on a fixed synthetic validation batch the model never trains on — measured across the full 10-epoch / 60,000-pair run:

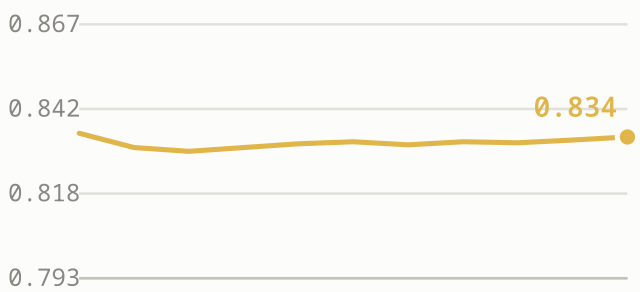
Held-out separation

pos-mean − neg-mean · climbs steadily



Pairwise rank accuracy

P(pos ranked above neg) · flat



Separation climbs steadily (real signal) — but pairwise rank accuracy, arguably the metric that matters more, stays flat at the random-init baseline. Training is pushing negative-pair scores down on average without becoming better at ranking any *specific* pair correctly. Reported as-is, not smoothed over.

TOTAL PAIRS TRAINED ON (ALL RUNS)

78,750

FULL RUN WALL TIME

7h 20m

10 epochs × 200 images × 30 pairs

MEASURED THROUGHPUT

0.44 s / pair

training (gradient); 155 pairs/s at inference

§4 Training time, if we scale further

Extrapolated from measured throughput (0.44 s/pair), not estimated in advance.

SCENARIO	TOTAL PAIRS	EST. TIME
Medium run — 15 epochs × 50 img × 12 pairs	9,000	1h 01m (actual)
Full run — 10 epochs × 200 img × 30 pairs	60,000	7h 20m (actual)
Double the full run — 20 epochs	120,000	~14.7 h
Triple the full run — 30 epochs	180,000	~22 h
Exhaustive — every patch position, 1 epoch	~168,000	~20.5 h

Plug in your own numbers — e.g. "if I only use 80 images, how long":

TIME CALCULATOR · BUILT FROM MEASURED THROUGHPUT (0.44 S/PAIR)

Training images used: 200

out of 200 clean images available

Epochs: 10

Pairs per image: 30

6/6 pos-neg split at 12; scales the same way

ESTIMATED TRAINING TIME

60,000 total pairs · 200 images/epoch × 10 epochs × 30 pairs/image

7h 20m

§5 Current status against the 5-step plan

✓ **1 · Train & sanity-check**

Done — plus the architecture bug found and fixed along the way, which wasn't in the original plan.

… **2 · Test on real images, per-category F1/IoU**

In progress. One early data point (image 001_F, no post-processing): F1 = 0.032, IoU = 0.016 — recall 0.81 but precision 0.017 (63,346 false-positive pixels vs. 1,065 true positives). Concerning, but it's one image — the full stratified 10-image benchmark hasn't run yet.

3 **3 · Rotation-aware offset voting**

Not started. Will add a dtheta bin to the offset vote, with a flag to fall back to translation-only.

4 **4 · Grid search (sim-threshold, block-window, min-offset-votes)**

Not started — blocked on step 2's baseline numbers.

5 **5 · 5-panel visualization tool**

Not started.

Generated from real run logs and checkpoint diagnostics — no figures on this page are illustrative placeholders.

`checkpoints_full/swaptest_qnn_epoch10.npy · code/cmfd_swaptest_pipeline.py`